

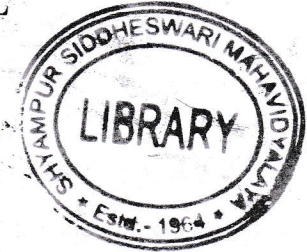
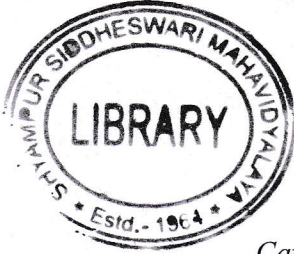
2022

COMPUTER SCIENCE — GENERAL

Paper : DSE-A-2

(Operations Research)

Full Marks : 50

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer **question no. 1** and **any four** questions from the rest.1. Answer **any five** questions :

2×5

- What is Basic feasible solution of a linear programming problem?
- Define unbalanced assignment problem.
- What is Slack variable?
- What is zero-sum game?
- What do you mean by Principle of duality?
- What is Degeneracy Problem?
- Give the formal definition of a transportation problem.

2. (a) Solve the following Assignment problem :

	A	B	C
1	120	100	80
2	80	90	110
3	110	140	120

(b) Give an example of a real-life application where this problem can be used.

8+2

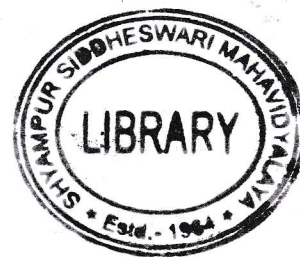
3. (a) Solve the following Transportation problem to find the initial basic feasible solution.

				a_j
	2	7	4	5
	3	3	1	8
	5	4	7	7
	1	6	2	14
b_j	7	9	18	

(b) When does a transportation problem become unbalanced?

8+2

Please Turn Over



4. (a) Solve the following L.P.P. using simplex method :

$$\text{Max } Z = 5x + 3y$$

$$\text{Subject to } 3x + 5y \leq 15$$

$$5x + 2y \leq 10$$

$$x, y \geq 0$$

- (b) When is surplus variable used?

8+2

5. (a) Write the dual of the following LPP.

$$\text{Maximize } Z = 2x_1 + x_2 - 3x_3 + 5x_4$$

$$\text{Subject to } x_1 + 7x_2 + 3x_3 + 7x_4 \leq 46$$

$$3x_1 - x_2 + x_3 + 2x_4 \leq 8$$

$$2x_1 + 3x_2 - x_3 + x_4 \leq 10$$

- (b) Solve graphically,

$$\text{Maximize } Z = x_1 + \frac{x_2}{2}$$

$$\text{Subject to } 3x_1 + 2x_2 \leq 12$$

$$5x_1 \leq 10$$

$$x_1 + x_2 \leq 8$$

$$-x_1 + x_2 \geq 4$$

4+6

6. Find the optimum strategies for A and B and the value of the game.

6+4

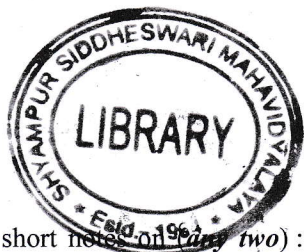
		B			
A		1	7	2	4
		0	3	7	8
		5	2	6	10

7. (a) Find the initial basic feasible solution of the following transportation problem using North-West corner method.

	A ₁	B ₁	C ₁	D ₁	E ₁	Supply
Origin A	2	11	10	3	7	4
B	1	4	7	2	1	8
C	3	9	4	8	12	9
Requirement	3	3	4	5	6	

- (b) Show that assignment problem is a special case of transportation problem.

8+2



(3)

X(5th Sm.)-Computer Sc.-G/DSE-A-2/CBCS

8. Write short notes on *any two* :

5×2

- (a) Saddle point of Game theory
 - (b) Branch and bound technique
 - (c) Critical Path Method (CPM)
 - (d) Unbalanced Transportation problem.
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